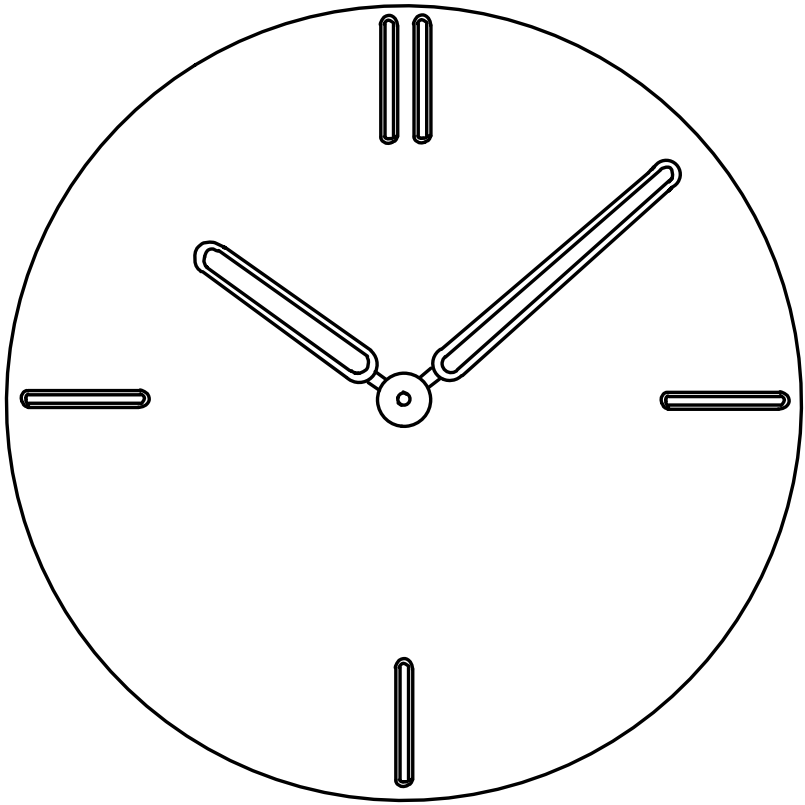
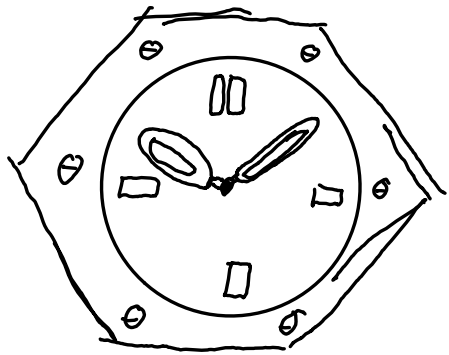
6 core beliefs:

- I'm inherently worth of infinite
  ↳ Inherent for glory
- Every interactions is a lesson if I let them
- Self-centeredness causes fear
- Mind is a tool as much as your body
- Problem is how you view a problem
- X Failure
  ✓ Feedback





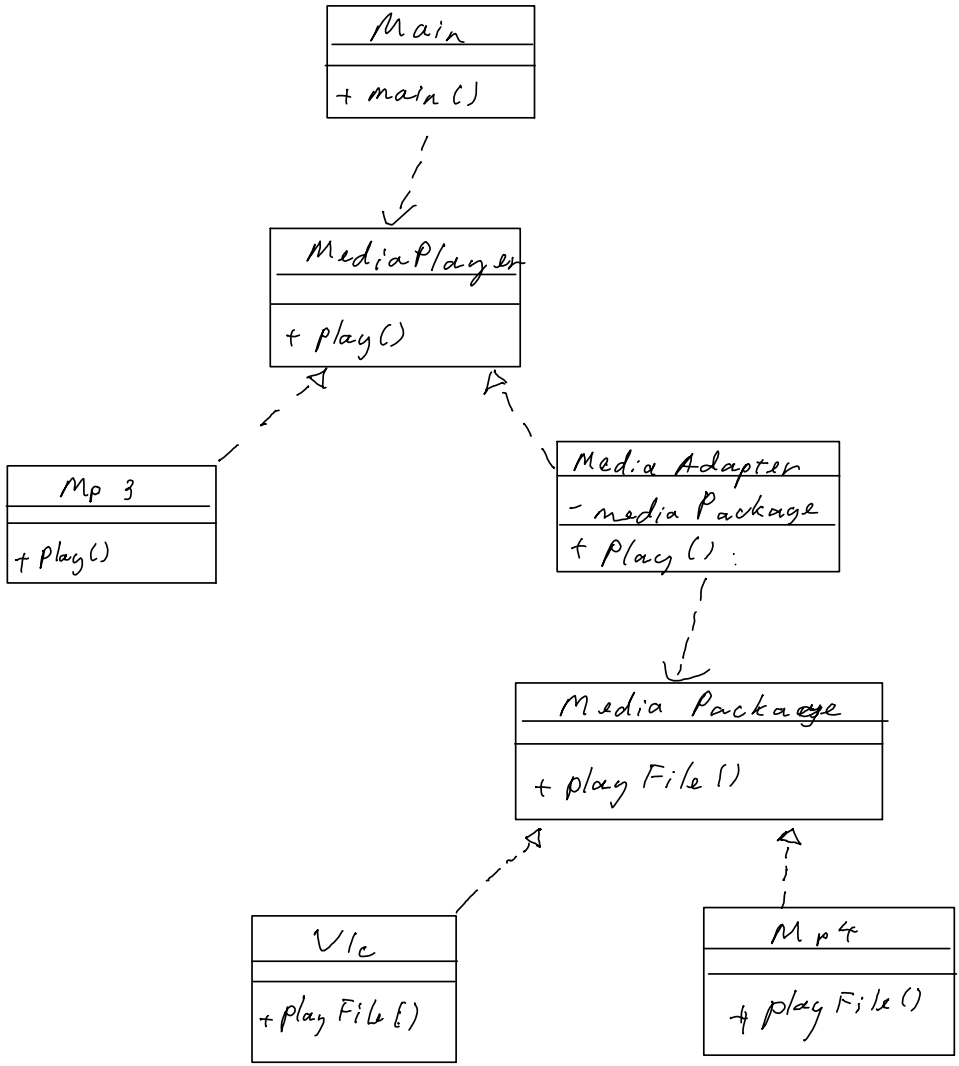
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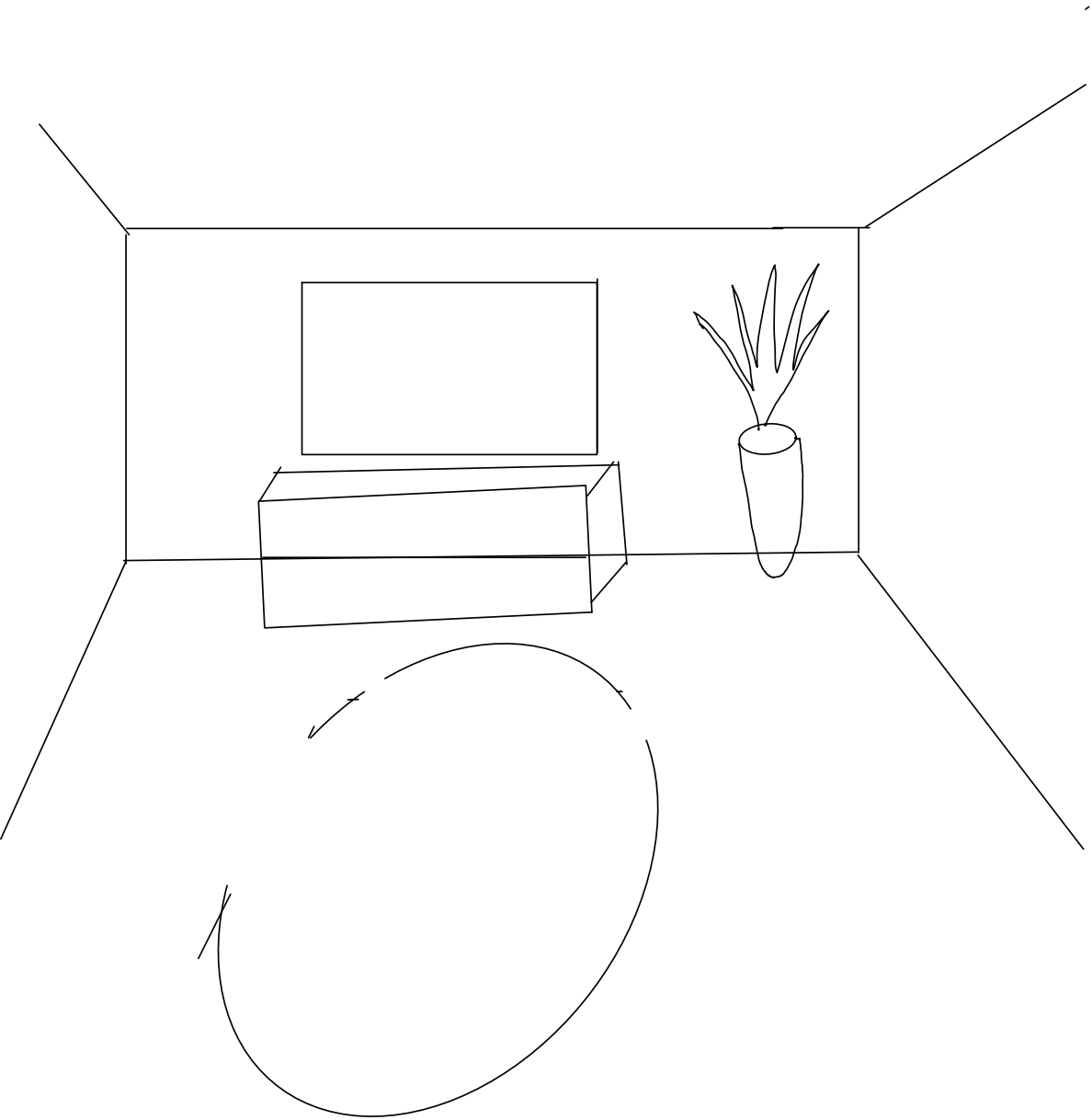
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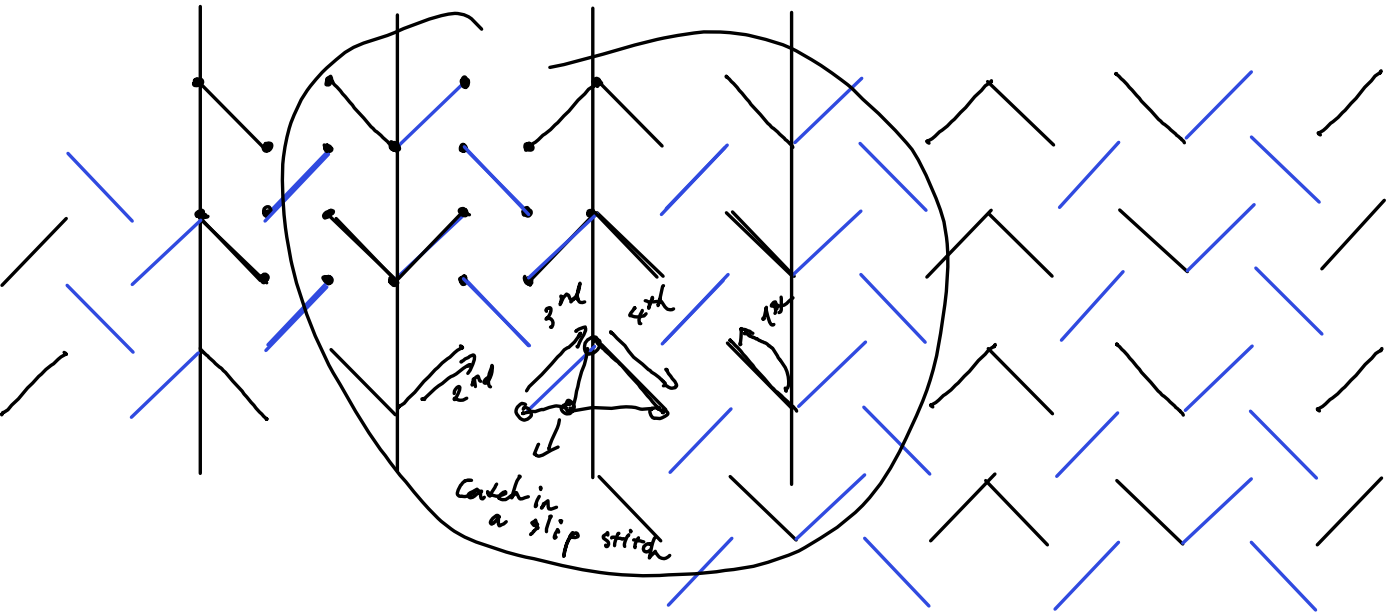
899999

900000

Trịnh Hồng Anh - 20235896






3rd
4th
1st
2nd
Catch in a slip stitch

Shoulder: ⟷ Ball and socket

- Flexion
- Extension
- Abduction
- Adduction
- Internal Rotation
- External Rotation
- horizontal abduction
- horizontal adduction
- Circum duction
- Scapular
  - protraction
  - retraction
  - elevation
  - depression
  - upward rotation
  - downward rotation

Elbow

- flexion
- extension
- pronation
- supination

Wrist

- flexion
- extension
- radial deviation
- Ulnar deviation
- circum duction

Fingers (index → pinky)
- flexion
- Extension (MCP/PIP/DIP)
- abduction
- adduction (MCP only)

Thumb
- Flexion
- Extension
- abduction
- adduction
- opposition
- reposition

Ankle
- Dorsiflexion
- plantarflexion
- Inversion
- Eversion
- Pronation
- Supination

Hip ⟶ Ball and socket
- Flexion
- Extension
- Abduction
- Adduction
- Internal Rot
- External Rot
- Circumduction

Knee
- flexion
- extension
- internal rot
- external rot

Sets :

$G = \{1, \ldots, N_G\}$

→ Ground devices

$S = \{1, \ldots, N_S\}$

→ Swarms

$A = \{1, \ldots, N_A\}$

→ Access Points

I) UAV mobility

1) Swarm center: $q_s[t] = [x_s[t], y_s[t], H_s]^T$

2) UAV pos : $q_{s,u}[t] = q_s[t] + \delta_{s,u}[t]$ (relative cord to)

3) Formation around center: $\sum_{u=1}^{N_s} \delta_{s,u}[t] = 0$

4) Formation Region (Limit relative cord)

$$\delta_{s,u}[t] \in D_s$$

5) Max Swarm Center Speed (slot duration)

$$\| q_s[t+1] - q_s[t] \|_2 \leq V_s^{max} \Delta$$

6) Swarm Center must in region: $q_s[t] \in Q_s$

7) Limit Speed for each UAV

$$\| q_{s,u}[t+1] - q_{s,u}[t] \|_2 \leq V_{s,u}^{max} \Delta$$

8) Min distance for UAVs in same swarm

$$\| q_{s,u}[t] - q_{s,v}[t] \|_2 \geq D_s^{min} = 0,5\ m$$

9) Min distance for swarms

$$\| q_s[t] - q_{s'}[t] \|_2 \geq D_{s,s'}^{safe} = 150\ m$$

II) Beamforming Model

10) Node position: $\tilde{q}_i[t] = \begin{cases} \bar{q}_i & i \in G \cup A \\ q_i[t] & i \in S \end{cases}$

11) Node distance: $d_{i,j}[t] = \left\| \tilde{q}_i[t] - \tilde{q}_j[t] \right\|_2$

12) Direction vector: $d_{s,j}[t] = \dfrac{\tilde{q}_j[t] - \tilde{q}_s[t]}{|d_{s,j}[t]|}$

13) Steering vector of swarm s to node j: $a_{s,j}[t] = \begin{bmatrix} e^{i\frac{2\pi}{\lambda}\delta_{s,1}[t]^T d_{s,j}[t]} \\ \dots \\ e^{i\frac{2\pi}{\lambda}\delta_{s,N_s}[t]^T d_{s,j}[t]} \end{bmatrix}$

14) Excitation weight: $0 \le I_{s,v,j}[t] \le 1$

$$\|I_{s,j}[t]\|_2^2 = 1 \qquad \left(I_{s,j}[t] = \begin{vmatrix} I_{s,1,j}[t] \\ \dots \\ I_{s,N_s,j}[t] \end{vmatrix} \right)$$

$\Rightarrow$ Main tain total transmit power fixed

15) Array factor

$$F_{s,j}(\theta,\phi,t) = \sum_{v=1}^{N_s} I_{s,v,j}[t]\, e^{i\frac{2\pi}{\lambda}\delta_{s,v}[t]^T d(\theta,\phi)}$$

$\Rightarrow$ "If swarm transmit to j, how power rec'd will the signal be?"

16) Array Response: $F^r_{s,j}[t] = \underbrace{I_{s,j}[t]^T}_{\text{weight}^T}\ \underbrace{a_{s,j}[t]}_{\text{steering vector}}$

17) Beamforming Gain:

$$G^{CB}_{s,j}[t] = 4\pi\eta_s \frac{|F^r_{s,j}[t]|^2 \ \ |w_s(\theta_{s,j},\phi_{s,j})|^2}{\int_0^{2\pi}\int_0^{\pi} |F_{s,j}(\theta,\phi,t)|^2 |w_s(\theta,\phi)|^2 \sin\theta \, d\theta \, d\phi}$$

$\eta_s \rightarrow$ array efficiency

$|F^r|^2 \rightarrow$ gain during hitting receiver

integral $\iint \rightarrow$ normalization

III) Channel

18) Air-to-Air channel gain: $h^{AA}_{i,j}[t] = \beta_0 \, d_{i,j}[t]^{-\alpha_{AA}}$

19) Ground-to-Air / Air-to-Ground elevation angle: $\vartheta_{ij}[t] = \frac{180}{\pi} \arcsin\left(\frac{H_{i,j}[t]}{d_{i,j}[t]}\right)$

20) Line-of-Sight (LoS) Probability: $P^{LoS}_{i,j}[t] = \frac{1}{1 + k_1 e^{-k_2(\vartheta_{i,j}[t] - k_1)}}$

$k_1, k_2 \rightarrow$ environment-dependent

21) Ground-Air gain: $h^{GA}_{i,j}[t] = \frac{\beta_0 \, d_{i,j}[t]^{-\alpha_{GA}}}{P^{LoS}_{i,j}[t]\,\eta_{LoS} + (1 - P^{LoS}_{i,j}[t])\,\eta_{NLoS}}$

$\eta_{LoS} \longrightarrow$ additional attenuation factors under LoS
$\eta_{NLoS} \longrightarrow$ ——————————— NLoS

22) Generic channel: $h_{i,j}[t] = \begin{cases} h^{AA}_{i,j}[t] \\ h^{GA}_{i,j}[t] \end{cases}$

23) Effective antenna gain: $G_{i,j}[t] = \begin{cases} G^{cB}_{i,j}[t] & , i \in S \\ 1 & , i \in G \end{cases}$

24) Achievable rate: $C_{i,j}[t] = B_{i,j} \log_2\left(1 + \underbrace{\frac{P_i G_{i,j}[t] \, h_{i,j}[t]}{\sigma^2_{i,j}}}_{SN}\right)$

25) Active links: $E[t] = \{(i,j) \in E : C_{i,j}[t] \geq C^{min}_{i,j}\}$

$C^{min} = 0.5$ Mbps

IV) Energy

26) Horizontal UAV speed : $$V_{s,u}[t] = \frac{\| q_{s,u}^{xy}[t+1] - q_{s,u}^{xy}[t] \|_2}{\Delta}$$

27) Propulsion power :

$$P_{s,u}^{fly}(v) = \underbrace{P_0 \left(1 + \frac{3v^2}{U_{tip}^2}\right)}_{\text{blade profile powers}} + \underbrace{\frac{1}{2} d_0 \rho s A r v^3}_{\text{parasitic / drag}} + \underbrace{P_I \left(\sqrt{1 + \frac{v^4}{4 v_0^4}} - \frac{v^2}{2 v_0^2}\right)^{1/2}}_{\text{induced powers}}$$

blade profile powers → profile power

parasitic / drag

induced powers → induced

28) Energy of UAV :

$$E_{s,u}^{fly} = \underbrace{\sum_{t=1}^{T-1} P_{s,u}^{fly}(V_{s,u}[t])\,\Delta}_{\text{flight energy}} + \underbrace{\frac{1}{2} m_{s,u} \left[(V_{s,u}^{after})^2 - (V_{s,u}^{ini})^2 \right]}_{\text{kinetic energy } \Delta} + \underbrace{m_{s,u}\, g \left(z_{s,u}[T] - z_{s,u}[1] \right)}_{\text{potential energy } \Delta}$$

29) Total UAV energy :

$$E^{fly} = \sum_{s \in S} \sum_{u=1}^{N_s} E_{s,u}^{fly}$$

30) Energy constraint : $E^{fly} \leq E_{th} = 40\, N_s N_s \; kJ$

V) TRAFFIC + QUEUE

31) Data available : $$O_g[t] = Q_g[t] + A_g[t]$$

32) Total delivered data : $$r_g[t] = \sum_{a \in A} r_{g,a}[t]$$

33) Delivered data cannot exceed available : $$0 \le \sum_{a \in A} r_{g,a}[t] \le O_g[t]$$

34) Queue update : $$Q_g[t+1] = Q_g[t] + A_g[t] - \sum_{a \in A} r_{g,a}[t]$$

VI) Multi-commodity flow

35) Flow conservation

$$\sum_{j:(v,j)\in E} f_{v,j}^{g,a}[t] - \sum_{i:(i,v)\in E} f_{i,v}^{g,a}[t] = \begin{cases} r_{g,a}[t] & , v = g \\ -r_{g,a}[t] & , v = a \\ 0 & , \text{otw} \end{cases}$$

$\Rightarrow$ data in = data out

36) Effective capacity

$$\tilde{C}_{i,j}[t] = \begin{cases} C_{i,j}[t] & C_{i,j}[t] \ge C_{i,j}^{min} \\ 0 \end{cases}$$

37) Link capacity constraint

$$\sum_g \sum_a f_{i,j}^{g,a}[t] \le \tilde{C}_{i,j}[t] \quad \Delta$$

38) Cummulative Queue :

$$Q[t+1;t] = Q_g^{ini} + \sum_{\tau=1}^{t} A_g[\tau] - \sum_{\tau=1}^{t} \sum_{a \in A} r_{g,a}[\tau]$$

39) Data causality

$$\sum_{\tau=1}^{t} \sum_{a \in A} r_{g,a}(\tau) \le Q_g^{ini} + \sum_{\tau=1}^{t} A_g[\tau]$$

$\Rightarrow$ All data sent $\le$ data existed

VII) OBJECTIVE

$$J(\Gamma, \Psi, I, F, r)$$

$$= \underbrace{\sum_t \sum_g \sum_a w_g \, r_{g,a}[t]}_{\text{weighted-reward for delivered data}} - \underbrace{\lambda_Q \sum_\tau \sum_g w_g \, Q_g[t+1;r]}_{\text{queue penalty: if queue} \uparrow \Rightarrow \text{penalty} \uparrow}$$

VIII) JOINT OPTIMIZATION

$$\max J(\Gamma, \Psi, I, F, r)$$

- $\Gamma = \{ q_s[t] \} \rightarrow$ swarm-center trajectories
- $\Psi = \{ \delta_{s,v}[t] \} \rightarrow$ individual UAV formations
- $I = \{ I_{s,j}[t] \} \rightarrow$ beamforming weights
- $F = \{ f_{i,j}^{g,a}[t] \} \rightarrow$ routing flows
- $r = \{ r_{g,a}[t] \} \rightarrow$ delivered data

